

1 Pre-planning

1.1 METHODOLOGY

The development of the Skill Barter System will follow the Incremental Software Development Methodology. This model divides the project into smaller, manageable increments that are developed, tested, and refined in successive cycles. Each increment adds functionality to the system, ensuring continuous progress and early feedback. This approach suits the project well since it allows flexibility in requirements and iterative improvements based on user testing.

Model Overview

- **Waterfall Model**

The Waterfall Model, introduced by Winston W. Royce in 1970, follows a simple step-by-step development process. The phases include Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. Each phase is completed before moving to the next one. This model is best for projects where requirements are clearly defined from the beginning and do not change frequently.

Advantages:

- Fixed timeline with well-defined phases.
- Easy to manage since each phase has clear deliverables.
- Good for small projects with predictable requirements.

Disadvantages:

- It is difficult to handle unexpected changes during development.
- Testing happens late, which may increase cost if major errors appear.
- Less flexibility and limited user feedback during early stages.

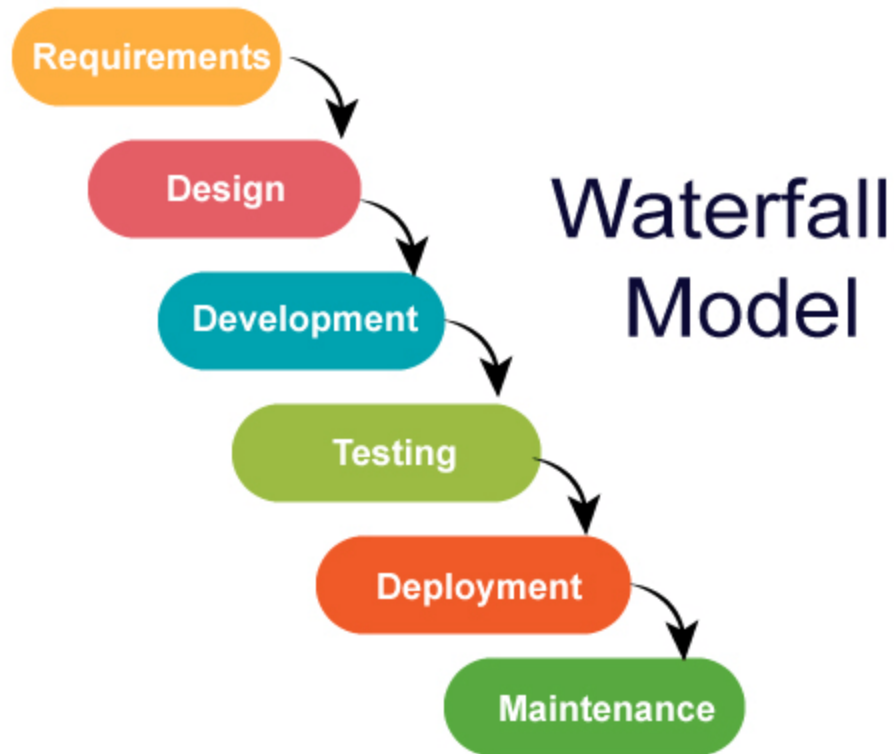


figure 1: Waterfall model

- **Prototype Model**

The Prototype Model focuses on building a quick prototype (or demo version) of the system before the final one is developed. This helps users understand what the system will look like and gives feedback early. The cycle includes Requirement Gathering, Quick Design, Prototype Development, Testing, and Refinement until the system meets user expectations.

Advantage:

- Encourages user involvement and feedback.
- Helps visualize the system early in development.
- Increase customer satisfaction because users see progress.

Disadvantages:

- Increase time and cost due to repeated refinements.
- Create confusion between the prototype and the final system.
- Lead to incomplete documentation.

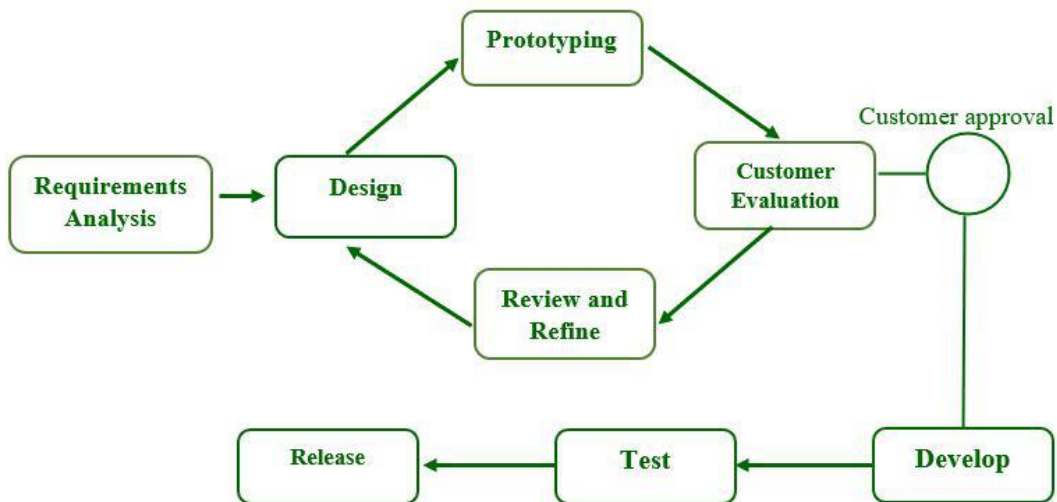


figure 2: Prototype model

• Spiral Model

The Spiral Model combines features of both the Waterfall and Prototype models. It is used for large or complex projects that need continuous improvement and risk management. The process repeats in cycles called *spirals*, where each loop includes Planning, Risk Analysis, Engineering, Testing, and Evaluation.

Advantages:

- Focus on risk management and problem-solving.
- Suitable for complex and long-term projects.
- Allows continuous refinement and customer input.

Disadvantages:

- Costly and time-consuming for small projects.
- Hard to manage due to complexity.
- Need experienced developers for risk assessment.

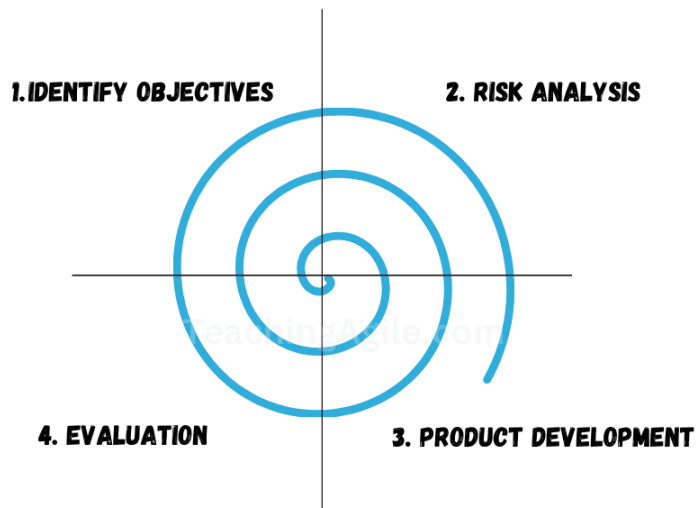
SPIRAL MODEL IN SOFTWARE DEVELOPMENT

figure 3: Spiral model

1.2 Chosen Methodology

The incremental model in software engineering divides a project into smaller modules, or increments, developed and integrated gradually. This iterative approach enables frequent testing, quick feedback, and early detection (Sachan, 2024).

The Incremental Model is a great fit for this project because it breaks the work into smaller steps, making development and integration easier. Each part is tested and improved before moving on, which helps create a more stable and unified system. This method also allows for regular feedback and makes it easier to adjust to changes, ensuring the final product meets user's needs and is strong and reliable.

Advantages of the Incremental Model:

- **Early Delivery:** The basic functions like registration, login and skill management were created during the initial phases and the system was able to generate visible outputs within a short period of time.
- **Flexibility:** The methodology enabled easy change to be made in the process of development. As an illustration, testing and feedback were used to enhance features like session handling and chat system.
- **Risk Reduction:** Mistakes and problems were detected during the initial stages of every increment, and the chances of significant failures in the subsequent stages were minimized.
- **Improved Quality:** Testing of each module was done separately followed by integration, which guaranteed better quality and reliability of the system.
- **Efficient Resource Utilization:** The development was done in small sections that could be handled easily and time and effort were easily allocated.
- **Improved Knowledge of System Requirements:** The requirements were more evident as the project went on, and it was possible to make improvements and refinements in subsequent increments.
- **Simplified Project Management:** Planning, monitoring and controlling the project was easier because the system was broken into smaller increments.
- **Scalability:** New modules like premium membership and payment integration were easily added without interfering with the current modules.

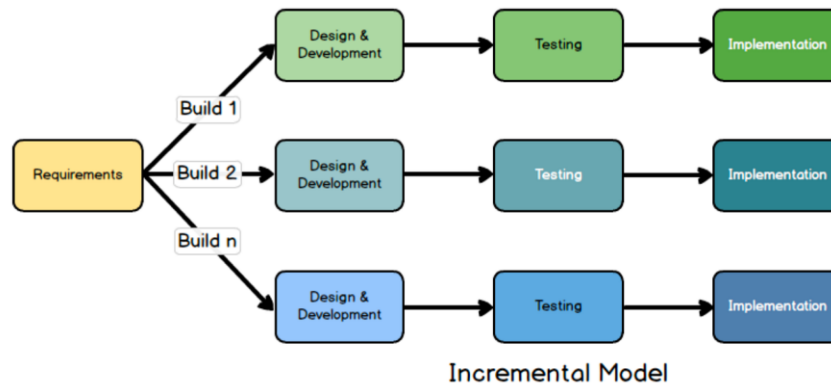


figure 4: Incremental Model

1.3 Incremental Breakdown

The development of the Skill Barter System was divided into multiple increments, where each increment focused on specific functionalities of the system.

- Increment 1 (Project Setup and System Design)

Tasks:

- Requirement Analysis: Research related journals and research papers.
- Comparison of other existing systems.
- System problems and solutions identified.
- Simple diagram drawing (Use Case, Flowchart), Complex diagram development (ERD, DFD, Sequence Diagram)
- Setting up environment (Laravel, database, tools)
- Wireframe design for user interface

- Increment 2 (User Authentication and Profile Management)

Tasks:

- Creation and configuration of databases.
- User registration functionality
- Login system implementation
- Email verification system
- Role management (Student, Teacher, Admin)
- Creation and updating profiles
- Basic dashboard setup

- Increment 3 (Skill Management and Request System)

Tasks:

- Development and management skills.
- Acquire teaching and learning skills.
- Search and skill browsing.
- Implementation of the session request system.
- Accept and reject request feature.
- Request status tracking: Blending of users and skills.

- Increment 4 (Session, Chat and Advanced Features)

Tasks:

- Creation of sessions upon acceptance of request.
- Status of sessions (pending, ongoing, completed)
- Real-time communication through a chat system.
- Resource sharing option (upload and download of files)
- Feedback and rating system.
- Notification system
- Premium membership feature
- Khalti payment integration.

- Increment 5 (Admin Panel, Testing and Finalization)

Tasks:

- Admin dashboard development
- User management (activate, deactivate, delete)
- Teacher approval system
- Admin skill management
- Observation and orders
- System testing and debugging
- Black-box testing
- Integration testing
- Performance testing
- Final documentation preparationmethodologiesmethodologySelected